

All
About
Pizza!



PIZZA SALES DATA ANALYSIS USING SQL

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HELLO !

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Is**

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**A Portfolio
Project for
Business
Intelligence
and Data
Analytics**

ABOUT PROJECT

The Pizza Sales Data Analysis Using SQL project focuses on analyzing a pizza restaurant's sales data to extract meaningful business insights. The project uses SQL to manage, query, and analyze data stored in a relational database. By working with multiple related tables, it demonstrates how SQL can be used to answer real-world business questions and support data-driven decision-making.

The analysis covers key performance indicators (KPIs) such as total revenue, total orders, best-selling pizzas, customer ordering patterns, sales by category and size, and peak ordering hours. Advanced SQL techniques, including joins, aggregate functions, Common Table Expressions (CTEs), and window functions, are used to perform comprehensive data analysis.

This project helps develop practical SQL skills while showcasing how database analysis can improve business performance by identifying popular products, sales trends, and revenue opportunities. It is designed as a portfolio project for aspiring Data Analysts, SQL Developers, and Business Intelligence professionals.



Project Objectives



- Analyze pizza sales performance.
- Calculate total revenue and total orders.
- Identify the best-selling pizzas.
- Analyze sales by category and pizza size.
- Determine peak ordering hours.
- Calculate cumulative revenue over time.
- Rank top-performing pizzas within each category.
- Generate business insights to support data-driven decision-making.





Technologies Used



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-
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MySQL
MySQL Workbench
SQL
CSV Dataset
GitHub



Database Tables

orders –

Stores order date and time.

order_details –

Stores pizzas ordered and quantities.

pizzas –

Contains pizza sizes and prices.

pizza_types –

- Contains pizza names, categories, and ingredients.



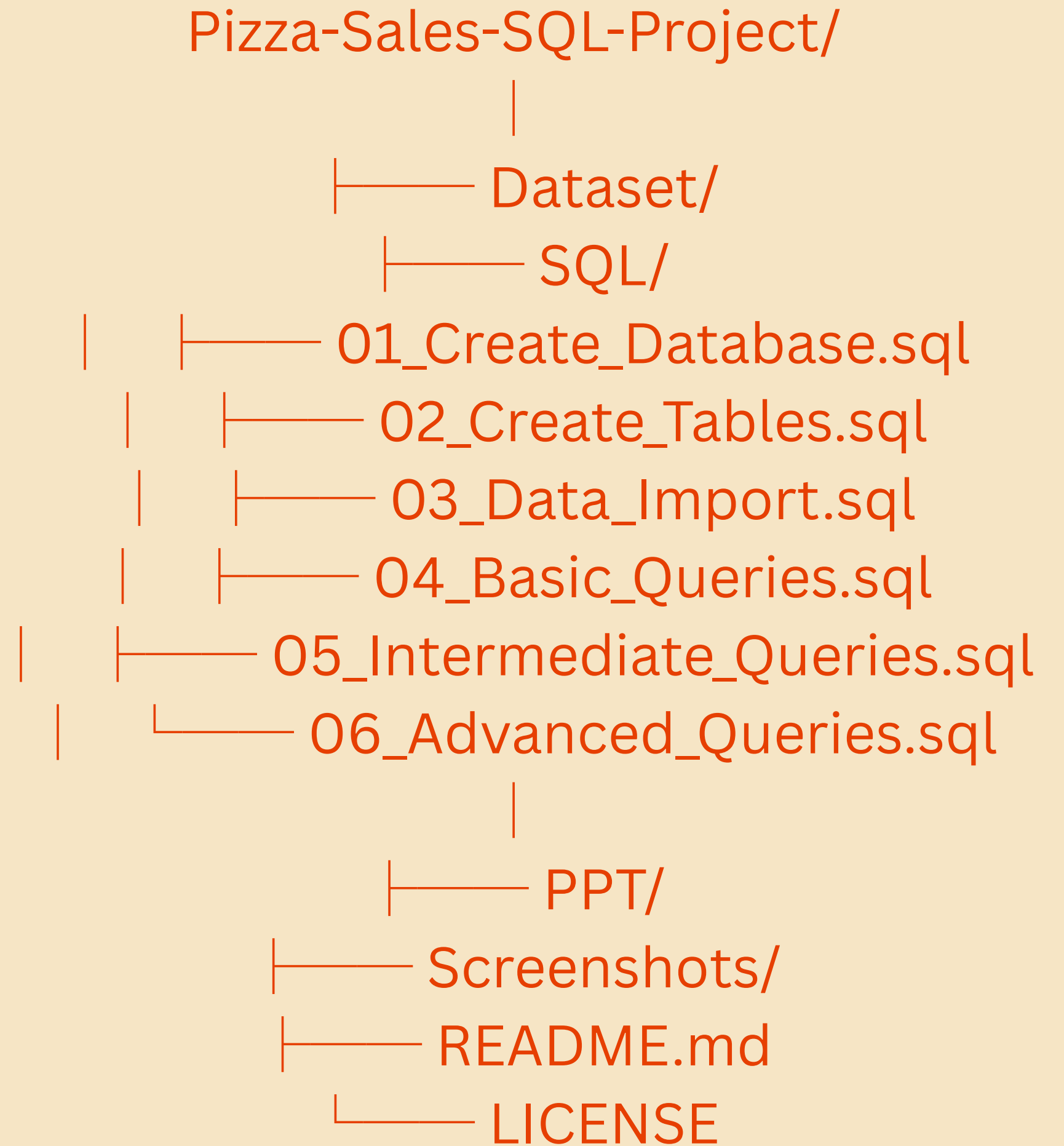
SQL Concepts Covered

- **CREATE DATABASE**
- **CREATE TABLE**
- **Primary & Foreign Keys**
- **INNER JOIN**
- **GROUP BY**
- **ORDER BY**
- **Aggregate Functions**
- **Subqueries**
- **Common Table Expressions (CTEs)**
- **Window Functions**
- **Ranking Functions**
- **Date & Time Functions**





Project Structure



Retrieve the total number of orders placed.



```
Query 1 | SQL File 1 | orders | SQL File 2 | orders_details | pizzas
-- Retrieve the total number of orders placed.
SELECT
    COUNT(order_id) AS total_orders
FROM
    orders;
```

Result Grid  Filter Rows:			
	order_date	revenue	cumulative_revenue
▶	2015-01-01	2713.85	2713.85
	2015-01-02	2731.9	5445.75
	2015-01-03	2662.4	8108.15
	2015-01-04	1755.45	9863.6
	2015-01-05	1139.3	11002.9

Calculate the total revenue generated from pizza sales.



```
1  -- Calculate the total revenue generated from pizza sales.  
2  • select  
3    sum(orders_details.quantity * pizzas.price) as total_sales  
4  from orders_details join pizzas  
5  on pizzas.pizza_id = orders_details.pizza_id;  
6
```

Result Grid	
	total_sales
▶	11002.9

Identify the most common pizza size ordered.

```
1  -- Identify the most common pizza size ordered.  
2  • select pizzas.size, count(orders_details.order_details_id) as order_count  
3  from pizzas join orders_details  
4  on pizzas.pizza_id=orders_details.pizza_id  
5  group by pizzas.size order by order_count desc ;
```



Result Grid			Filter
	size	order_count	
▶	L	274	
	M	194	
	S	170	
	XL	6	

Join the necessary tables to find the total quantity of each pizza category ordered.



```
-- Join the necessary tables to find the total quantity of each pizza category ordered.
```

```
SELECT
```

```
    pizza_types.category,  
    SUM(orders_details.quantity)as quantity  
from pizza_types join pizzas  
on pizza_types.pizza_type_id = pizzas.pizza_type_id  
join orders_details  
on orders_details.pizza_id = pizzas.pizza_id  
group by pizza_types.category order by quantity desc;
```

Result Grid



Filter R

	category	quantity
	Classic	190
	Supreme	159
	Chicken	156
	Veggie	152

Determine the distribution of orders by hour of the day.



```
1  -- Determine the distribution of orders by hour of the day
2  SELECT
3  HOUR(order_time) AS order_hour,
4  COUNT(order_id) AS total_orders
5  FROM orders
6  GROUP BY order_hour
7  ORDER BY order_hour;
```

Result Grid			Filter Rows:
	order_hour	total_orders	
▶	10	3	
	11	402	
	12	891	
	13	817	
	14	543	
	15	492	
	16	650	
	17	860	
	18	797	
	19	689	
	20	552	

Calculate the percentage contribution of each pizza type to total revenue.

```
1  -- Calculate the percentage contribution of each pizza type to total revenue.
2
3  •  SELECT
4      pizza_types.name,
5      ROUND(SUM(orders_details.quantity * pizzas.price),2) AS revenue,
6      ROUND(
7          SUM(orders_details.quantity * pizzas.price) * 100 /
8          (
9              SELECT SUM(orders_details.quantity * pizzas.price)
10             FROM orders_details
11             JOIN pizzas
12             ON orders_details.pizza_id = pizzas.pizza_id
13          ),
14      2) AS revenue_percentage
15  FROM pizza_types
16  JOIN pizzas
17  ON pizza_types.pizza_type_id = pizzas.pizza_type_id
18  JOIN orders_details
19  ON pizzas.pizza_id = orders_details.pizza_id
20  GROUP BY pizza_types.name
21  ORDER BY revenue_percentage DESC;
```

	name	revenue	revenue_percent
	The California Chicken Pizza	563.25	5.12
	The Spicy Italian Pizza	535	4.86
	The Five Cheese Pizza	481	4.37
	The Classic Deluxe Pizza	472	4.29
	The Greek Pizza	421.5	3.83
	The Pepperoni Pizza	412.5	3.75
	The Four Cheese Pizza	398.8	3.62
	The Mexicana Pizza	395	3.59
	The Southwest Chicken Pizza	388.5	3.53
	The Sicilian Pizza	386	3.51
	The Italian Capocollo Pizza	373.5	3.39
	The Hawaiian Pizza	369.25	3.36
	The Prosciutto and Arugula...	344.25	3.13
	The Napolitana Pizza	339.5	3.09
	The Chicken Alfredo Pizza	326.25	2.97
	The Spinach Supreme Pizza	294.75	2.68
	The Vegetables + Vegetabl...	289.75	2.63
	The Spinach Pesto Pizza	282.25	2.57
	The Pepper Salami Pizza	281.75	2.56
	The Big Meat Pizza	264	2.4
	The Chicken Pesto Pizza	225	2.04
	The Italian Vegetables Pizza	214.75	1.95
	The Spinach and Feta Pizza	209.25	1.9
	The Green Garden Pizza	180.25	1.64
	The Pepperoni, Mushroom,...	161.5	1.47
	The Soppressata Pizza	153.5	1.4
	The Mediterranean Pizza	141	1.28

Analyze the cumulative revenue generated over time.



```
1  -- Analyze the cumulative revenue generated over time.
2
3  •  SELECT
4      order_date,
5      revenue,
6      SUM(revenue) OVER (ORDER BY order_date) AS cumulative_revenue
7  FROM
8  (
9      SELECT
10         orders.order_date,
11         ROUND(SUM(orders_details.quantity * pizzas.price), 2) AS revenue
12     FROM orders
13     JOIN orders_details
14         ON orders.order_id = orders_details.order_id
15     JOIN pizzas
16         ON orders_details.pizza_id = pizzas.pizza_id
17     GROUP BY orders.order_date
18 ) AS daily_revenue;
```

	order_date	revenue	cumulative_revenue
►	2015-01-01	2713.85	2713.85
	2015-01-02	2731.9	5445.75
	2015-01-03	2662.4	8108.15
	2015-01-04	1755.45	9863.6
	2015-01-05	1139.3	11002.9

THANK
YOU

